4-phenylbutyrate class and bile-acid chemical chaperones

Scheda A01_pba_class · Obiettivo: Recupero ampio e duraturo · Track B (specifico di variante o lesione) · Stato: controversa · F3 ipotesi favorevole dal bersaglio, non provata in WWOX

Data dei dati: 2026-10-28 (registro dei bisogni 2026-10-27; versione del catalogo 2026-10-28T23:40:16Z). **Perimetro:** encefalopatia epilettica e dello sviluppo legata a WWOX (WWOX-DEE), classe del genotipo di riferimento; solo letteratura e basi dati pubbliche; nessun dato individuale. Materiale di ricerca per la discussione con il team clinico curante: non è consiglio medico, non è una raccomandazione di trattamento, non è una probabilità di beneficio.

Motivazione

WOREE (Q230P-class missense) as target. Origin contexts: STXBP1, SLC6A1, GABRA1 and GAT-1 variants in cells, worm, fly, zebrafish and mouse (NOT WWOX EVIDENCE). Bridge limit: one precedent of the lot is a soluble cytosolic client (Munc18-1) where the class raised the mutant, but no enzyme and no WWOX system was tested; wild type is raised as well.

Bisogni affrontati: Salvare una proteina WWOX destabilizzata da una variante missenso del dominio SDR.

Evidenza, per tipo

- ▲Precedente in un'altra malattia variant protein expression and rescue: increase – The class rescued the fully retained variant and NOT the partly trafficked G443D; the rescued variants recover part of wild-type capacity with altered affinity; wild-type capacity is raised too.
- ▲Precedente in un'altra malattia wild-type protein abundance under the class: increase – The class raises wild-type protein in every target of the lot: the effect is not variant-specific.
- ▲Precedente in un'altra malattia functional rescue: no_change – MEASURED NO RESCUE (behaviour unchanged). Encoded NO_CHANGE: a reported null, not an unmeasured cell. In the same paper glycerol, an osmolyte-type chemical chaperone, did rescue G443D in vitro: the partly trafficked branch was outside the reach of this class, …
- ▲Precedente in un'altra malattia movement deficit and ictal-like events: no_change – MEASURED NO RESCUE in null alleles, protein not measured, bath concentration bounded by larval toxicity, assay separation of controls unestablished; INCONCLUSIVE for a protein-expressing missense allele. Encoded NO_CHANGE.
- ▲Precedente in un'altra malattia mature-to-immature ratio and function: increase – The HDAC-inhibitor comparator raised the total amount of immature protein with a small, variant-dependent function gain (the authors read non-specific accumulation;
- ▲Precedente in un'altra malattia total variant protein, 4-phenylbutyrate arm: increase – Total mutant protein about 1.8 fold of its own untreated control (three experiments, significance mark printed); wild type rose about 1.8 fold as well, so the gain is not mutant-specific; transgene mRNA under treatment was not measured. (PMID 30266908)

- ▲ Precedente in un'altra malattia total variant protein, class-I HDAC-inhibitor comparator (butyrate) at the same applied level: no_change – Mutant total protein about 1.05 of untreated control, no significance mark (three experiments); the text lists only the phenylbutyrate class and two osmolytes as stabilising. On wild type the same comparator gave a scattered, unmarked value near 1.95. (PMID 30266908)
- ▲ Precedente in un'altra malattia total variant protein, two osmolyte chemical chaperones with no HDAC activity (sorbitol, trehalose): increase – Mutant total protein about 1.8 (sorbitol) and about 2.35 (trehalose) fold of control, significance marks printed; also raised in yeast by a second assay. Both act at high osmolyte levels, so an osmotic effect is not separated from chaperoning; (PMID 30266908)
- ▲ Precedente in un'altra malattia synaptotagmin-1 antibody uptake (function of the same protein), 4-phenylbutyrate arm: increase – Mutant total protein sits at about 4 to 8 percent of wild type in four of five variants and rises only about 1.05 to 1.65 fold with the class; the soluble fraction roughly doubles (soluble pool about 2 to 4 fold, reader arithmetic); (PMID 30266908)
- ▲ Precedente in un'altra malattia liver mutant enzyme protein and plasma phenylalanine (function read-out): increase – Enzyme protein rose toward wild type with acetylated HSP90 up; plasma phenylalanine fell about 44 percent from sham yet stayed about three times wild type; two sham animals against three treated; (PMID 41274966)
- *Osservazioni registrate sui bisogni collegati: solo contesto, possono riguardare altri interventi.*
- ● Esperimento WWOX in un sistema DEE *osservazione sul bisogno:* Patient-derived fibroblast line homozygous for p.Gln230Pro: WWOX transcript is comparable to reference lines, but no WWOX band is detected on one western membrane at one exposure (confirmed on a second harvest of the same line). (PMID 29808465)
- ■ Biologia WWOX in un altro contesto *osservazione sul bisogno:* A different SDR missense (P252A), stably over-expressed in non-neural thyroid-carcinoma cell lines (CAL-62, BCPAP): the P252A protein is much lower than wild type while its mRNA is not decreased; a cycloheximide chase shows faster turnover; (PMID 41124647)

Nella scheda non è registrato alcun esito di questo intervento in un sistema WWOX.

Evidenza contraria

- Wild-type protein is raised too.
- Null zebrafish negative (protein not measured).
- Partly trafficked variant not rescued by the class (glycerol rescued it in vitro).
- Rationale withdrawn once in this repository.
- Class-I HDAC action does not reproduce the soluble mutant gain, but that does not exclude HDAC6 or transcription-linked effects.
- Observed gains also apply to wild type and are small against a deficit of tens of fold.
- *Bisogno:* Q230P is not P47T and not P252A; no read-across between missense alleles (MECHANISM_TRANSFER_FIREWALL).
- *Bisogno:* Abundance restored without function would be neutral at best (CLAIM 030).
- *Bisogno:* N-11: 4-phenylbutyrate/TUDCA act on an ER-chaperone rationale and WWOX is cytosolic. N-12: HSP70 co-induction could stabilise an insoluble protein (adjacent-protein precedent: stabilisation gave insoluble protein without activity).

Limiti

- Associazione non è modulazione, e modulazione non è recupero; un precedente di un'altra malattia non è un risultato WWOX.
- I tipi di evidenza restano separati e non sono mai accorpati; fonti condivise non sono conferme indipendenti.
- Le categorie di accesso sono evidenza di accesso trovata nelle fonti (prevalentemente USA) alle date di recupero (non approvazione, non idoneità per una persona, non sicurezza): con prescrizione, sconosciuto o non verificato.
- Il testo delle schede può citare riferimenti interni del repository (per esempio «CLAIM 031», «A-VGB-5», «T1», «DL-…», «verifica cieca», «nessuna ricevuta nel registro»): non servono alla lettura e non sono consultabili fuori dal repository. Solo gli identificativi nella tabella delle fonti sono pubblici.

Limiti scritti nella scheda (testo originale in inglese):

- Soluble monomeric client with a dominant-negative mechanism, or ER-retained membrane client, versus a recessive cytosolic SDR enzyme; CMV-promoter plasmid overexpression in cell lines, or lentiviral expression near wild-type level in neurons (authors' statement;
- Technical comparison is not efficacy, safety or experimental validation.
- Sources and receipt references are retained, not independently verified.
- Shared sources are not independent confirmations; scores are not pooled.

Evidenza mancante e prossimo esperimento

Dato mancante: Where the Q230P protein is lost: never made, made and cleared, made and insoluble, made and epitope-invisible, or truncated. No fractionation, inhibitor, synthesis or half-life datum exists for the exact variant..

Passo successivo: Fate first: three-fraction (soluble, pellet, total) plus an orthogonal detection and a route-inhibitor arm on the exact variant, with transgene mRNA measured in the same wells under treatment; then a wild-type titration of the chosen function read-out as a ruler; then a lever panel that separates three classes (a class-I HDAC inhibitor without HDAC6 activity, a pan-HDAC or selective HDAC6 inhibitor, HDAC-free bile acid and osmolytes with an osmotic control), with the arm-outcome predictions registered in both directions (design in missense_recovery_it5_p8_20261029/README.md section 8). No abundance gain is to be read as rescue before the ruler exists.

Fonti (identificativi pubblici)

- PMID 29808465 `https://pubmed.ncbi.nlm.nih.gov/29808465/` A novel missense variant in the SDR domain of the WWOX gene leads to complete loss of WWOX protein with early-onset epileptic encephalopathy and severe developmental delay. [Neurogenetics, 2018] (observation source in the need record)
- PMID 30266908 `https://pubmed.ncbi.nlm.nih.gov/30266908/` Mechanism-based rescue of Munc18-1 dysfunction in varied encephalopathies by chemical chaperones. [Nat Commun, 2018] (source of an intervention assertion)
- PMID 41124647 `https://pubmed.ncbi.nlm.nih.gov/41124647/` Genetic and Functional Evidence Links Germline Biallelic Inactivating Variants in WWOX to Histological

Mixed-Type Thyroid Cancer. [Adv Sci (Weinh), 2026] (observation source in the need record)

- PMID 41274966 `https://pubmed.ncbi.nlm.nih.gov/41274966/` Therapeutic effect of histone deacetylase 6 inhibitor for a mouse model of phenylketonuria. [Sci Rep, 2025] (source of an intervention assertion)
- ChEMBL CHEMBL1746 `https://www.ebi.ac.uk/chembl/compound_report_card/CHEMBL1746/` (identità del composto)

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